

Year 7 · Chapter 3 · Whole numbers: answers (teacher copy)

3.01 Rounding and estimation (booklet p. 3)

Examples (model these live)

E1 Round 6 482 to the nearest hundred.

hundreds digit 4, next digit 8: round up
· $\approx 6\,500$

E2 Estimate 389×52 .
 $\approx 400 \times 50 \cdot \approx 20\,000$

E3 27 students each pay \$48. Estimate the total.

$\approx 30 \times 50 \cdot \approx \$1\,500$

Questions (39 on today's skill, 15 from earlier lessons)

1 1 500

2 9 000

3 2 000

4 800

5 700

6 2 000

7 270

8 900

9 6 ^{1.05}

10 9 000

11 3 000

12 550

13 $\angle AVK$ or $\angle KVA$ ^{2.01}

14 $\approx 30 \times 10 \cdot \approx 300$

15 1 000

16 3 000

17 $a = 95$ ^{2.10}

18 $\approx 50 + 70 \cdot \approx 120$

19 $\approx 2\,000 \div 70 \cdot \approx 29$

20 -5 ^{1.02}

21 88 000

22 $\approx 8\,000 - 3\,000 \cdot \approx 5\,000$

23 largest: 9 · 9, -2, -10, -25
^{1.03}

24 $\approx 40 + 60 \cdot \approx 100$

25 $\approx 6\,000 \div 40 \cdot \approx 150$

26 $\approx 8\,000 \div 50 \cdot \approx 160$

27 96 000

28 -2 ^{1.01}

29 $\approx 7\,000 - 3\,000 \cdot \approx 4\,000$

30 $18 - (-14) \cdot = 18 + 14 \cdot =$

32°C ^{1.10}

31 $\approx 80 \times 70 \cdot \approx 5\,600$

32 $\approx 30 \times 10 \cdot \approx \300

33 $= -15 - 3 \cdot = -18$ ^{1.08}

34 $\approx 5\,000 \div 80 \cdot \approx 63$

35 $= 14 - 31 \cdot = -17$ ^{1.04}

36 21 000

37 $\approx 30 \times 40 \cdot \approx \$1\,200$

38 $\approx 800 \div 30 \cdot \approx 27$

39 $360^\circ - 315^\circ \cdot = 45^\circ$ ^{2.02}

40 $\approx 700 \times 20 = 14\,000$ · No:
about 10 times too big (10
576)

41 $2x + 142 = 180 \cdot 2x = 38 \cdot x$
 $= 19$ ^{2.11}

42 250

43 $\approx 20 \times 20 \cdot \approx \400

44 5 ^{1.07}

45 $\approx 800 \div 20 \cdot \approx 40$

46 $\approx 700 \div 30 \cdot \approx 23$

47 449

48 58° ^{2.07}

49 $\approx 40 \times 30 \cdot \approx \$1\,200$

50 $= 6 \times 3 \cdot = 18$ ^{1.06}

51 $\approx 8\,000 - 5\,000 \cdot \approx 3\,000$

52 $\approx 400 \div 30 \cdot \approx 13$

53 111° ^{2.04}

54 849

3.02 Multiplication (booklet p. 5)

Examples (model these live)

E1 34×6

$= 30 \times 6 + 4 \times 6 \cdot = 180 + 24 \cdot = 204$

E2 56×300

$= 56 \times 3 \times 100 \cdot = 168 \times 100 = 16\,800$

E3 243×17

$243 \times 10 = 2\,430 \cdot 243 \times 7 = 1\,701 \cdot 2$
 $430 + 1\,701 = 4\,131$

Questions (43 on today's skill, 17 from earlier lessons)

1 100

2 750

3 360

4 $= 10 \times 5 + 7 \times 5 \cdot = 50 + 35 \cdot$
 $= 85$

5 88

6 32 000

7 640

8 $= 30 \times 3 + 7 \times 3 \cdot = 90 + 21 \cdot$
 $= 111$

9 40 ^{3.01}

10 72

11 72 000

12 280

13 7 700

14 180

15 $= 30 \times 3 + 1 \times 3 \cdot = 90 + 3 \cdot$
 $= 93$

16 they are equal ^{2.11}

17 120

18 $= 70 \times 5 + 1 \times 5 \cdot = 350 + 5 \cdot$
 $= 355$

19 370

20 2 000 ^{3.01}

21 200

22 $= 40 \times 9 + 8 \times 9 \cdot = 360 + 72 \cdot$
 $= 432$

23 $= 70 \times 7 + 6 \times 7 \cdot = 490 +$

$42 \cdot = 532$

24 $= 200 \times 5 + 34 \times 5 \cdot = 1\,000$
 $+ 170 \cdot = 1\,170$

25 $\approx 8\,000 - 5\,000 \cdot \approx 3\,000$

^{3.01}

26 $= 8 \times 9 \times 1\,000 \cdot = 72\,000$

27 $42 \times 12 \cdot = 504$ stickers

28 119° ^{2.07}

29 $= 90 \times 6 + 5 \times 6 \cdot = 540 +$
 $30 \cdot = 570$

30 $= 300 \times 9 + 26 \times 9 \cdot = 2\,700$
 $+ 234 \cdot = 2\,934$

31 $= 6 \times 8 \times 1\,000 \cdot = 48\,000$

32 20 ^{3.01}

33 $25 \times 17 \cdot = 425$ cards

34 $\approx 9\,000 \div 30 \cdot \approx 300$ ^{3.01}

35 $= 70 \times 6 + 8 \times 6 \cdot = 420 +$
 $48 \cdot = 468$

36 $418 \times 20 = 8\,360 \cdot 418 \times 2$
 $= 836 \cdot 8\,360 + 836 = 9\,196$

37 largest: 2 · 2, -9, -15, -21
^{1.03}

38 $= 300 \times 7 + 93 \times 7 \cdot = 2\,100$
 $+ 651 \cdot = 2\,751$

39 380 ^{3.01}

40 $= 4 \times 4 \times 1\,000 \cdot = 16\,000$

41 $306 \times 30 = 9\,180 \cdot 306 \times 1$

$= 306 \cdot 9\,180 + 306 = 9\,486$

42 $= 100 \times 50 - 50 \cdot = 5\,000 -$
 $50 \cdot = 4\,950$

43 $\approx 7\,000 - 2\,000 \cdot \approx 5\,000$
^{3.01}

44 seats: $18 \times 29 = 522 \cdot 522 \times$
 $15 \cdot = \$7\,830$

45 5 ^{1.04}

46 $84 \div 3 = 28 \cdot \square = 2$

47 $241 \times 10 = 2\,410 \cdot 241 \times 5$
 $= 1\,205 \cdot 2\,410 + 1\,205 = 3\,615$

48 3 000 ^{3.01}

49 $= 100 \times 30 - 30 \cdot = 3\,000 -$
 $30 \cdot = 2\,970$

50 $\approx 9\,000 \div 20 \cdot \approx 450$ ^{3.01}

51 $= 100 \times 55 - 55 \cdot = 5\,500 -$
 $55 \cdot = 5\,445$

52 seats: $17 \times 17 = 289 \cdot 289 \times$
 $14 \cdot = \$4\,046$

53 $((-20) \times 2) \cdot = 400$ ^{1.09}

54 $356 \div 4 = 89 \cdot \square = 8$

55 50 ^{3.01}

56 $426 \times 20 = 8\,520 \cdot 426 \times 3$
 $= 1\,278 \cdot 8\,520 + 1\,278 = 9\,798$

57 $25 \times 20 \cdot = 500$ cards

58 $\approx 5\,000 - 3\,000 \cdot \approx 2\,000$
^{3.01}

59 $= 100 \times 52 - 52 \cdot = 5\,200 -$
 $52 \cdot = 5\,148$

60 parallel ^{2.06}

3.03 Division (booklet p. 7)

Examples (model these live)

E1 $72 \div 8$

$8 \times 9 = 72 \cdot = 9$

E2 $4\,500 \div 100$

move each digit two places right · =
45

E3 $852 \div 4$

$8 \div 4 = 2, 5 \div 4 = 1 \text{ r } 1 \cdot 12 \div 4 = 3 \cdot =$
213

Questions (43 on today's skill, 16 from earlier lessons)

1 7

2 51

3 12

4 10

- 5 33
6 9
7 3
8 25
9 15 000 ^{3.02}
10 8
11 9
12 36
13 9
14 10
15 8
16 4
17 parallel ^{2.06}
18 $= 320 \div 8 + 8 \div 8 \cdot = 40 + 1 \cdot = 41$
19 11
20 8
21 4 000 ^{3.01}
22 11
- 23 $= 80 \div 8 + 8 \div 8 \cdot = 10 + 1 \cdot = 11$
24 short division, left to right $\cdot = 163$
25 $= 8 \times 5 \times 1\,000 \cdot = 40\,000$ ^{3.02}
26 $5 \times 28 = 140 \cdot = 28 \text{ r } 4$
27 $= 540 \div 9 + 36 \div 9 \cdot = 60 + 4 \cdot = 64$
28 63° ^{2.08}
29 short division, left to right $\cdot = 151$
30 $5 \times 51 = 255 \cdot = 51 \text{ r } 3$
31 short division, left to right $\cdot = 216$
32 $6 \times 34 = 204 \cdot = 34 \text{ r } 2$
33 8 000 ^{3.01}
34 $= 180 \div 6 + 18 \div 6 \cdot = 30 + 3 \cdot = 33$
35 $28 \times 16 \cdot = 448$ cards ^{3.02}
- 36 $271 \div 9 = 30 \text{ r } 1 \cdot 30$ each, 1 left over
37 short division, left to right $\cdot = 182$
38 $\alpha = 150 \cdot$ corresponding angles, $\parallel$ lines ^{2.07}
39 $3 \times 145 = 435 \cdot = 145 \text{ r } 2$
40 31 000 ^{3.01}
41 $450 \div 7 = 64 \text{ r } 2 \cdot 64$ each, 2 left over
42 $70 \div 9 = 7 \text{ r } 7 \cdot 8$ vans (one more for the rest)
43 $= 70 \times 6 + 4 \times 6 \cdot = 420 + 24 \cdot = 444$ ^{3.02}
44 $\square = 25 \times 8 \cdot \square = 200$
45 3 ^{1.02}
46 short division, left to right $\cdot = 1\,026$
47 $165 \div 6 = 27 \text{ r } 3 \cdot 27$ each, 3 left over
- 48 9 000 ^{3.01}
49 $97 \div 7 = 13 \text{ r } 6 \cdot 14$ vans (one more for the rest)
50 $= 100 \times 5 + 23 \times 5 \cdot = 500 + 115 \cdot = 615$ ^{3.02}
51 $73 \div 6 = 12 \text{ r } 1 \cdot 13$ vans (one more for the rest)
52 $\square = 41 \times 5 \cdot \square = 205$
53 $4 \times 30^\circ = 120^\circ \cdot$ obtuse ^{2.03}
54 short division, left to right $\cdot = 2\,466$
55 6 000 ^{3.01}
56 $238 \div 5 = 47 \text{ r } 3 \cdot 47$ each, 3 left over
57 $7 \times 71 = 497 \cdot = 71 \text{ r } 2$
58 $= 6 \times 7 \times 1\,000 \cdot = 42\,000$ ^{3.02}
59 $79 \div 6 = 13 \text{ r } 1 \cdot 14$ vans (one more for the rest)

3.04 Divisibility tests (booklet p. 9)

Examples (model these live)

E1 Is 4 527 divisible by 3?

$4 + 5 + 2 + 7 = 18 \cdot$ yes: 18 is divisible by 3

E2 Is 7 316 divisible by 4?

last two digits: 16 $\cdot$ yes: $16 \div 4 = 4$

E3 Is 5 742 divisible by 6?

even, and digit sum 18 $\cdot$ yes: divisible by 2 and 3

Questions (40 on today's skill, 16 from earlier lessons)

- 1 no
2 17; no
3 395
4 no
5 20; no
6 895
7 yes
8 23; no
9 64 ^{3.03}
10 16; no
11 875
12 no
13 $180^\circ - 31^\circ \cdot = 149^\circ$ ^{2.09}
14 digit sum: 25 $\cdot$ no: 25 is not divisible by 9
15 15; yes
16 455
17 340 ^{3.02}
18 no
19 digit sum: 20 $\cdot$ no: 20 is not divisible by 9
- 20 last two digits: 24 $\cdot$ yes: 24 is divisible by 4
21 $\approx 8\,000 \div 70 \cdot \approx 114$ ^{3.01}
22 even? yes; digit sum 27 $\cdot$ yes: divisible by 2 and 3
23 last three digits: 804 $\cdot$ no: $804 \div 8 = 100 \text{ r } 4$
24 18 ^{1.06}
25 digit sum: 26 $\cdot$ no: 26 is not divisible by 9
26 last two digits: 55 $\cdot$ no: 55 is not divisible by 4
27 last two digits: 64 $\cdot$ yes: 64 is divisible by 4
28 even? no; digit sum 26 $\cdot$ no: not divisible by both 2 and 3
29 6 ^{3.03}
30 last three digits: 794 $\cdot$ no: $794 \div 8 = 99 \text{ r } 2$
31 $= 2 \times 2 \times 1\,000 \cdot = 4\,000$ ^{3.02}
32 digit sum: 30 $\cdot$ no: 30 is not divisible by 9
- 33 $5 + 1 + 6 = 12 \cdot 12 + \square$ must be a multiple of 3 $\cdot \square = 0, 3, 6, 9$
34 12 apart, half is 6 $\cdot 6$ right of -4: 2 ^{1.02}
35 last two digits: 47 $\cdot$ no: 47 is not divisible by 4
36 3 000 ^{3.01}
37 even? no; digit sum 21 $\cdot$ no: not divisible by both 2 and 3
38 $1 + 6 + 4 = 11 \cdot 11 + \square$ must be a multiple of 9 $\cdot \square = 7$
39 even, digit sum 21, last two digits 56 $\cdot 2, 3, 4, 6$
40 $= 400 \div 8 + 32 \div 8 \cdot = 50 + 4 \cdot = 54$ ^{3.03}
41 No. Check the last two digits: 10 is not divisible by 4.
42 0 ^{1.05}
43 1 000
44 $2 + 6 + 1 = 9 \cdot 9 + \square$ must be a multiple of 3 $\cdot \square = 0, 3, 6, 9$
- 45 $= 40 \times 7 + 2 \times 7 \cdot = 280 + 14 \cdot = 294$ ^{3.02}
46 even, digit sum 12, last two digits 20 $\cdot 2, 3, 4, 5, 6, 8, 10$
47 even, digit sum 27, last two digits 96 $\cdot 2, 3, 4, 6, 9$
48 No. Check the last two digits: 58 is not divisible by 4.
49 $c + 20 + 30 = 180 \cdot c = 130$ ^{2.05}
50 1 000
51 39 ^{3.03}
52 $5 + 1 + 1 = 7 \cdot 7 + \square$ must be a multiple of 9 $\cdot \square = 2$
53 last three digits: 163 $\cdot$ no: $163 \div 8 = 20 \text{ r } 3$
54 $= 80 \times 5 + 1 \times 5 \cdot = 400 + 5 \cdot = 405$ ^{3.02}
55 even, digit sum 21, last two digits 28 $\cdot 2, 3, 4, 6, 8$
56 155° ^{2.04}

3.05 Long division (booklet p. 11)

Examples (model these live)

E1 $672 \div 16$

$67 \div 16 = 4 \text{ r } 3 \cdot$ bring down 2: $32 \div 16 = 2 \cdot = 42$

E2 $851 \div 23$

$85 \div 23 = 3 \text{ r } 16 \cdot$ bring down 1: $161 \div 23 = 7 \cdot = 37$

E3 500 people go in buses that hold

45. How many buses?
 $500 \div 45 = 11 \text{ r } 5 \cdot 12$ buses

Questions (39 on today's skill, 15 from earlier lessons)

- 1 115
2 4
3 2 r 4
4 108
5 4
6 7 r 10
- 7 126
8 3
9 62 ^{3.03}
10 5
11 4 r 14
12 54
- 13 $\angle WFA$ or $\angle AFW$ ^{2.01}
14 $68 \div 12 = 5 \text{ r } 8 \cdot$ bring down: $84 \div 12 = 7 \cdot = 57$
15 8
16 8 r 4
17 11; no ^{3.04}
- 18 126
19 $56 \div 14 = 4 \text{ r } 0 \cdot$ bring down: $0 \div 14 = 0 \cdot = 40$
20 divide, multiply, subtract, bring down $\cdot = 245$
21 $\approx 9\,000 \div 90 \cdot \approx 100$ ^{3.01}

- 22 $17 \times 24 = 408 \div = 24 \text{ r } 15$
 23 $77 \div 19 = 4 \text{ r } 1 \cdot$ bring down: $19 \div 19 = 1 \cdot = 41$
 24 $= 300 \times 7 + 21 \times 7 \cdot = 2\,100 + 147 \cdot = 2\,247$ ^{3.02}
 25 divide, multiply, subtract, bring down $\cdot = 160$
 26 $24 \times 41 = 984 \cdot = 41 \text{ r } 14$
 27 divide, multiply, subtract, bring down $\cdot = 205$
 28 $24 \times 28 = 672 \cdot = 28 \text{ r } 22$
 29 $180^\circ - 50^\circ \cdot = 130^\circ$ ^{2.09}
- 30 $52 \div 22 = 2 \text{ r } 8 \cdot$ bring down: $88 \div 22 = 4 \cdot = 24$
 31 $7\,000$ ^{3.01}
 32 $758 \div 21 \cdot = 36 \text{ r } 2 \cdot 36$ teams, 2 left over
 33 divide, multiply, subtract, bring down $\cdot = 205$
 34 7 ^{3.03}
 35 $15 \times 34 = 510 \cdot = 34 \text{ r } 4$
 36 even? no; digit sum 13 $\cdot$ no: not divisible by both 2 and 3 ^{3.04}
 37 $76 \div 12 = 6 \text{ r } 4 \cdot$ bring down: $48 \div 12 = 4 \cdot = 64$
- 38 $781 \div 26 \cdot = 30 \text{ r } 1 \cdot 30$ teams, 1 left over
 39 $671 \div 51 = 13 \text{ r } 8 \cdot 14$ buses
 40 $y = 180 - 55 = 125$ (straight) $\cdot x = 125$ (corresponding) ^{2.10}
 41 $4\,125 \div 15 \cdot = \$275$
 42 360 ^{3.02}
 43 $19 \times 32 + 18 \cdot = 626$
 44 $1\,083 \div 38 \cdot = 28 \text{ r } 19 \cdot 28$ teams, 19 left over
 45 $\approx 6\,000 - 2\,000 \cdot \approx 4\,000$ ^{3.01}
- 46 $778 \div 42 = 18 \text{ r } 22 \cdot 19$ buses
 47 $2\,717 \div 13 \cdot = \$209$
 48 $66 - 103 \cdot = -37$ dollars ^{1.10}
 49 $14 \times 29 + 8 \cdot = 414$
 50 $9\,000$ ^{3.01}
 51 $937 \div 25 \cdot = 37 \text{ r } 12 \cdot 37$ teams, 12 left over
 52 $24 \times 22 = 528 \cdot = 22 \text{ r } 18$
 53 no ^{3.04}
 54 $986 \div 45 = 21 \text{ r } 41 \cdot 22$ buses

3.06 Roman numerals (booklet p. 13)

Examples (model these live)

- E1 Write XLVII as a number.
XL = 40, VII = 7 $\cdot$ = 47
- E2 Write 94 in Roman numerals.
 $90 + 4 \cdot$ XC + IV = XCIV
- E3 Write 2026 in Roman numerals.
 $2000 + 20 + 6 \cdot$ MM + XX + VI = MMXXVI

Questions (40 on today's skill, 17 from earlier lessons)

- 1 **53**
 2 **XXXVI**
 3 **5**
 4 **55**
 5 **LXI**
 6 **10**
 7 **25**
 8 **XXVI**
 9 **4** ^{3.05}
 10 **LXVII**
 11 **50**
 12 **65**
 13 **2** ^{1.05}
 14 $200 + 60 + 4 \cdot$ CC + LX + IV = CCLXIV
 15 **XCIII**
 16 **1 000**
 17 **17**; no ^{3.04}
 18 **41**
 19 $200 + 60 + 3 \cdot$ CC + LX + III = CCLXIII
 20 $CD + LX + V \cdot 400 + 60 + 5 = 465$
- 21 $\approx 5\,000 \div 50 \cdot \approx 100$ ^{3.01}
 22 **XLVIII = 48, LXXXVI = 86** $\cdot$ larger: LXXXVI
 23 $700 + 10 + 9 \cdot$ DCC + X + IX = DCCXIX
 24 short division, left to right $\cdot = 151$ ^{3.03}
 25 $CM + XC + VII \cdot 900 + 90 + 7 = 997$
 26 **XLV = 45, XXVIII = 28** $\cdot$ larger: XLV
 27 $C + XX + I \cdot 100 + 20 + 1 = 121$
 28 **XCI = 91, LVII = 57** $\cdot$ larger: XCI
 29 $-1\,054$ ^{1.09}
 30 $700 + 80 + 7 \cdot$ DCC + LXXX + VII = DCCLXXXVII
 31 **730** ^{3.02}
 32 $1000 + 900 + 70 + 5 \cdot$ M + CM + LXX + V $\cdot$ = MCMLXXV
 33 $CM + IV \cdot 900 + 4 = 904$
 34 **7 r 11** ^{3.05}
 35 **LVI = 56, LXIII = 63** $\cdot$ larger: LXIII
- 36 even? yes; digit sum 14 $\cdot$ no: not divisible by both 2 and 3 ^{3.04}
 37 $300 + 30 \cdot$ CCC + XXX = CCCXXX
 38 $1000 + 800 + 90 + 4 \cdot$ M + DCCC + XC + IV $\cdot$ = MDCCCXCIV
 39 $58 + 15 = 73 \cdot$ = LXXIII
 40 $4 \times 30^\circ = 120^\circ \cdot$ obtuse ^{2.03}
 41 No. Only I, X and C can go before a larger letter (I before V or X, X before L or C, C before D or M). 95 = XCV.
 42 **6** ^{3.03}
 43 $1000 + 900 + 60 + 1 \cdot$ M + CM + LX + I $\cdot$ = MCMLXI
 44 $19 + 45 = 64 \cdot$ = LXIV
 45 **75 000** ^{3.01}
 46 No. Only I, X and C can go before a larger letter (I before V or X, X before L or C, C before D or M). 999 = CMXCIX.
- 47 $70 \div 18 = 3 \text{ r } 16 \cdot$ bring down: $162 \div 18 = 9 \cdot = 39$ ^{3.05}
 48 $53 + 14 = 67 \cdot$ = LXVII
 49 No. Only I, X and C can go before a larger letter (I before V or X, X before L or C, C before D or M). 99 = XCIX.
 50 **5** ^{1.03}
 51 $1000 + 800 + 10 \cdot$ M + DCCC + X $\cdot$ = MDCCCX
 52 **160** ^{3.02}
 53 **XCVIII = 98, LXX = 70** $\cdot$ larger: XCVIII
 54 $58 + 35 = 93 \cdot$ = XCIII
 55 no ^{3.04}
 56 No. Only I, X and C can go before a larger letter (I before V or X, X before L or C, C before D or M). 49 = XLIX.
 57 $= 80 \div 8 + 56 \div 8 \cdot = 10 + 7 \cdot = 17$ ^{3.03}

3.07 Index notation and powers (booklet p. 15)

Examples (model these live)

- E1 Write $3 \times 3 \times 3 \times 3$ in index form.
four 3s multiplied $\cdot$ = 3^4
- E2 Evaluate 2^5 .
 $= 2 \times 2 \times 2 \times 2 \times 2 \cdot = 32$
- E3 Evaluate $4^2 + 2^3$.
 $= 16 + 8 \cdot = 24$

Questions (40 on today's skill, 17 from earlier lessons)

- 1 **9^2**
 2 **9**
 3 $= 3 \times 3 \times 3 \cdot = 27$
 4 **4^3**
 5 **4**
 6 $= 10 \times 10 \times 10 \times 10 \cdot = 10\,000$
 7 **4^5**
 8 **144**
 9 **19 000** ^{3.02}
 10 **121**
- 11 $= 10 \times 10 \times 10 \cdot = 1\,000$
 12 **8^3**
 13 **b = 75** ^{2.05}
 14 $= 4 \times 4 \times 4 \times 4 \times 4 \cdot$ base 4, index 5
 15 **49**
 16 $= 4 \times 4 \times 4 \cdot = 64$
 17 **XCI** ^{3.06}
 18 **2^4**
 19 $= 7 \times 7 \times 7 \times 7 \times 7 \cdot$ base 7, index 5
 20 $= 5 \times 5 \times 5 \times 5 \cdot = 625$
- 21 $\approx 7\,000 \div 20 \cdot \approx 350$ ^{3.01}
 22 **$2^3 = 8, 3^2 = 9$ $\cdot$ larger: 3^2**
 23 $= 16 + 27 \cdot = 43$
 24 divide, multiply, subtract, bring down $\cdot = 119$ ^{3.05}
 25 $= 3 \times 3 \times 3 \cdot$ base 3, index 3
 26 $= 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \cdot = 256$
 27 $= 7 \times 7 \times 7 \cdot = 343$
 28 **$4^5 = 1024, 5^4 = 625$ $\cdot$ larger: 4^5**
 29 **7 105**
- 30 $= 9 + 8 \cdot = 17$
 31 **41** ^{3.03}
 32 $= 8 \times 8 \times 8 \times 8 \times 8 \cdot$ base 8, index 6
 33 $= 2^2 \times 5^1 \cdot = 4 \times 5 \cdot = 20$
 34 **12**; yes ^{3.04}
 35 $= 6 \times 6 \times 6 \cdot = 216$
 36 $= 3 \times 8 \times 1\,000 \cdot = 24\,000$ ^{3.02}
 37 **$2^5 = 32, 5^2 = 25$ $\cdot$ larger: 2^5**
 38 $= 2^2 \times 5^2 \cdot = 4 \times 25 \cdot = 100$

- 39 $3 \times 3 \times \dots$ until $81 \cdot n = 4$
 40 $= -25 \times (-1) \cdot = 25$ ^{1.06}
 41 10^6
 42 50 ^{3.06}
 43 No. $3^2 = 3 \times 3 = 9$, not 3×2 .
 44 $= 2^4 \times 5^2 \cdot = 16 \times 25 \cdot = 400$
 45 $96\ 000$ ^{3.01}
 46 $4 \times 4 \times \dots$ until $64 \cdot n = 3$
 47 $23 \times 24 = 552 \cdot = 24 \text{ r } 1$ ^{3.05}
 48 $3 \times 3 \times \dots$ until $243 \cdot n = 5$
 49 10^3
 50 $a = 35$ ^{2.10}
 51 No. $6^2 = 6 \times 6 = 36$, not 6×2 .
 52 4 ^{3.03}
 53 $= 2^3 \times 5^1 \cdot = 8 \times 5 \cdot = 40$
 54 $= 9 + 27 \cdot = 36$
 55 975 ^{3.04}
 56 $2 \times 2 \times \dots$ until $32 \cdot n = 5$
 57 $16 \times 7 \cdot = 112$ cards ^{3.02}

3.08 Square roots and cube roots (booklet p. 17)

Examples (model these live)

- E1 $\sqrt{64}$
 $8 \times 8 = 64 \cdot = 8$
 E2 $\sqrt[3]{125}$
 $5 \times 5 \times 5 = 125 \cdot = 5$
 E3 Between which two whole numbers is $\sqrt{40}$?
 $6^2 = 36$ and $7^2 = 49 \cdot$ between 6 and 7

Questions (44 on today's skill, 17 from earlier lessons)

- 1 5
 2 4
 3 10
 4 7
 5 2
 6 7
 7 11
 8 3
 9 16 ^{3.07}
 10 3
 11 4
 12 5
 13 8
 14 1
 15 9
 16 12
 17 29 ^{1.01}
 18 $10^2 = 100$ and $11^2 = 121 \cdot$ between 10 and 11
 19 4
 20 3
 21 XIII ^{3.06}
 22 $9^2 = 81$ and $10^2 = 100 \cdot$ between 9 and 10
 23 $2^2 = 4$ and $3^2 = 9 \cdot$ between 2 and 3
 24 $= 2 + 7 \cdot = 9$
 25 $\approx 5\ 000 \div 80 \cdot \approx 63$ ^{3.01}
 26 $= 5 \times 2 \cdot = 10$
 27 $8 \times 8 \times 8 = 512 \cdot = 8$
 28 divide, multiply, subtract, bring down $\cdot = 151$ ^{3.05}
 29 $10^2 = 100$ and $11^2 = 121 \cdot$ between 10 and 11
 30 $= 7 + 5 \cdot = 12$
 31 $= 5 + 2 \cdot = 7$
 32 $= 4 \times 4 \cdot = 16$
 33 acute ^{2.03}
 34 $10 \times 10 \times 10 = 1\ 000 \cdot = 10$
 35 510 ^{3.02}
 36 $11^2 = 121$ and $12^2 = 144 \cdot$ between 11 and 12
 37 $= \sqrt{25} \cdot = 5$
 38 $= 2 \times 2 \times 2 \times 2 \cdot = 16$ ^{3.07}
 39 $= 2 + 2 \cdot = 4$
 40 LXXVI = 76, LXVI = 66 $\cdot$ larger: LXXVI ^{3.06}
 41 $= 5 \times 8 \cdot = 40$
 42 $= \sqrt{289} \cdot = 17$
 43 $\sqrt{169} \cdot = 13$ cm
 44 $= 1 \times 1 \cdot = 1$ ^{1.06}
 45 $\sqrt[3]{8} \cdot = 2$ cm
 46 $5 \text{ r } 11$ ^{3.05}
 47 $\sqrt{(16 + 9)} = \sqrt{25} = 5 \cdot \sqrt{16} + \sqrt{9} = 7$: no
 48 $= \sqrt{225} \cdot = 15$
 49 short division, left to right $\cdot = 200$ ^{3.03}
 50 $\sqrt{144} \cdot = 12$ cm
 51 $= 36 + 27 \cdot = 63$ ^{3.07}
 52 $\sqrt{36} \cdot = 6$ cm
 53 $\sqrt[3]{125} \cdot = 5$ cm
 54 57° ^{2.08}
 55 $= \sqrt{100} \cdot = 10$
 56 15; yes ^{3.04}
 57 $6 \times 6 \times 6 = 216 \cdot = 6$
 58 $\sqrt{121} \cdot = 11$ cm
 59 44 ^{3.06}
 60 $\sqrt[3]{27} \cdot = 3$ cm
 61 $68 \div 18 = 3 \text{ r } 14 \cdot$ bring down: $144 \div 18 = 8 \cdot = 38$ ^{3.05}

3.09 Prime and composite numbers (booklet p. 19)

Examples (model these live)

- E1 List the factors of 18.
 $1 \times 18, 2 \times 9, 3 \times 6 \cdot = 1, 2, 3, 6, 9, 18$
 E2 Is 51 prime or composite?
 $51 = 3 \times 17 \cdot$ composite
 E3 List the primes between 20 and 40.
 test each odd number $\cdot = 23, 29, 31, 37$

Questions (40 on today's skill, 17 from earlier lessons)

- 1 composite
 2 $1 \times 18, 2 \times 9, 3 \times 6 \cdot 1, 2, 3, 6, 9, 18$
 3 4
 4 composite
 5 $1 \times 26, 2 \times 13 \cdot 1, 2, 13, 26$
 6 4
 7 prime
 8 $1 \times 10, 2 \times 5 \cdot 1, 2, 5, 10$
 9 25; no ^{3.04}
 10 $1 \times 35, 5 \times 7 \cdot 1, 5, 7, 35$
 11 6
 12 composite
 13 acute ^{2.03}
 14 composite: $81 = 3 \times 27$
 15 $1 \times 38, 2 \times 19 \cdot 1, 2, 19, 38$
 16 4
 17 2 ^{3.08}
 18 composite
 19 composite: $90 = 2 \times 45$
 20 41, 43, 47, 53, 59
 21 $\approx 9\ 000 \div 40 \cdot \approx 225$ ^{3.01}
 22 e.g. $34 = 3 + 31$
 23 composite: $76 = 2 \times 38$
 24 $= 5 \times 5 \times 5 \times 5 \cdot = 625$ ^{3.07}
 25 11, 13, 17, 19, 23, 29
 26 e.g. $28 = 5 + 23$
 27 71, 73, 79, 83, 89
 28 e.g. $18 = 5 + 13$
 29 6 ^{1.02}
 30 composite: $80 = 2 \times 40$
 31 $1\ 200$ ^{3.02}
 32 37
 33 23, 29, 31, 37
 34 LXXIX ^{3.06}
 35 e.g. $38 = 7 + 31$
 36 even? yes; digit sum 22 $\cdot$ no: not divisible by both 2 and 3 ^{3.04}
 37 composite: $55 = 5 \times 11$
 38 107
 39 No. $91 = 7 \times 13$, so it is composite.
 40 $10 - (-6) \cdot = 10 + 6 \cdot = 16^\circ\text{C}$ ^{1.05}
 41 2 and 13
 42 9 ^{3.08}
 43 $1 \times 92, 2 \times 46, 4 \times 23 \cdot 6: 1, 2, 4, 23, 46, 92$
 44 109
 45 short division, left to right $\cdot = 129$ ^{3.03}
 46 No. $119 = 7 \times 17$, so it is composite.
 47 $2^3 = 8, 3^2 = 9 \cdot$ larger: 3^2 ^{3.07}
 48 No. $51 = 3 \times 17$, so it is composite.
 49 2. Every other even number has 2 as a factor.
 50 11° ^{2.04}
 51 $1 \times 70, 2 \times 35, 5 \times 14, 7 \times 10 \cdot 8: 1, 2, 5, 7, 10, 14, 35, 70$
 52 3 ^{3.05}
 53 41
 54 e.g. $30 = 7 + 23$
 55 100 ^{3.06}
 56 No. $57 = 3 \times 19$, so it is composite.
 57 last three digits: $173 \cdot$ no: $173 \div 8 = 21 \text{ r } 5$ ^{3.04}

3.10 Prime factors (booklet p. 21)

Examples (model these live)

- E1 Write 36 as a product of primes.
 $36 = 4 \times 9 = 2 \times 2 \times 3 \times 3 \cdot = 2^2 \times 3^2$
 E2 Write 90 as a product of primes.
 $90 = 9 \times 10 = 3 \times 3 \times 2 \times 5 \cdot = 2 \times 3^2 \times 5$
 E3 Which number is $2^3 \times 5$?
 $= 8 \times 5 \cdot = 40$

Questions (40 on today's skill, 17 from earlier lessons)

- | | | | |
|---|--|---|---|
| 1 $= 27 \times 7 = 189$ | 19 $168 = 2 \times 2 \times 2 \times 3 \times 7 = 2^3 \times 3 \times 7$ | 33 $5 = 5, 8 = 2 \times 2 \times 2 \cdot 40 = 2^3 \times 5$ | 45 short division, left to right $\cdot = 206$ ^{3.03} |
| 2 $2^3 \times 3^3$ | 20 $6 = 2 \times 3, 10 = 2 \times 5 \cdot 60 = 2^2 \times 3 \times 5$ | 34 2 ^{3.09} | 46 Yes. 6's prime factors are all in it. |
| 3 2, 5 | 21 $\approx 6\,000 \div 40 \cdot \approx 150$ ^{3.01} | 35 $= 3 \times 3 \times 5 \times 5 = 225$ | 47 composite: $96 = 2 \times 48$ ^{3.09} |
| 4 $= 27 \times 5 = 135$ | 22 $= 2 \times 2 \times 2 \times 2 \times 3 = 48$ | 36 $2^5 = 32, 5^2 = 25 \cdot$ larger: 2^5 ^{3.07} | 48 $225 = 3^2 \times 5^2 = (3 \times 5)^2$, so $\sqrt{225} = 15$ |
| 5 2×3^2 | 23 $198 = 2 \times 3 \times 3 \times 11 = 2 \times 3^2 \times 11$ | 37 $192 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 3 = 2^6 \times 3$ | 49 Yes. 12's prime factors are all in it. |
| 6 2, 7 | 24 $= 9 \times 7 = 16$ ^{3.08} | 38 $520 = 2 \times 2 \times 2 \times 5 \times 13 = 2^3 \times 5 \times 13$ | 50 2 ^{1.04} |
| 7 $= 9 \times 7 = 63$ | 25 $9 = 3 \times 3, 4 = 2 \times 2 \cdot 36 = 2^2 \times 3^2$ | 39 $400 = 2^4 \times 5^2 = (2^2 \times 5)^2$, so $\sqrt{400} = 20$ | 51 $810 = 2 \times 3 \times 3 \times 3 \times 3 \times 5 = 2 \times 3^4 \times 5$ |
| 8 $2^3 \times 3$ | 26 $= 2 \times 2 \times 7 = 28$ | 40 small: $\angle YPR, \angle RPL$ · whole: $\angle YPL$ ^{2.01} | 52 11; no ^{3.04} |
| 9 $1 \times 34, 2 \times 17 \cdot 1, 2, 17, 34$ ^{3.09} | 27 $4 = 2 \times 2, 6 = 2 \times 3 \cdot 24 = 2^3 \times 3$ | 41 No. 2's prime factors are not all in it. | 53 $= 2 \times 2 \times 3 \times 5 = 60$ |
| 10 $2^2 \times 3^3$ | 28 $= 2 \times 3 \times 3 \times 5 = 90$ | 42 12 ^{3.08} | 54 $144 = 2^4 \times 3^2 = (2^2 \times 3)^2$, so $\sqrt{144} = 12$ |
| 11 2, 3 | 29 $-1\,440$ ^{1.09} | 43 $460 = 2 \times 2 \times 5 \times 23 = 2^2 \times 5 \times 23$ | 55 9^2 ^{3.07} |
| 12 $= 3 \times 7 = 21$ | 30 $171 = 3 \times 3 \times 19 = 3^2 \times 19$ | 44 $196 = 2^2 \times 7^2 = (2 \times 7)^2$, so $\sqrt{196} = 14$ | 56 No. 4's prime factors are not all in it. |
| 13 $180^\circ - 127^\circ = 53^\circ$ ^{2.09} | 31 180 ^{3.02} | | 57 $9 \times 9 \times 9 = 729 = 9$ ^{3.08} |
| 14 $189 = 3 \times 3 \times 3 \times 7 = 3^3 \times 7$ | 32 $684 = 2 \times 2 \times 3 \times 3 \times 19 = 2^2 \times 3^2 \times 19$ | | |
| 15 $2^2 \times 3^2$ | | | |
| 16 2, 3 | | | |
| 17 49 ^{3.07} | | | |
| 18 $= 4 \times 5 = 20$ | | | |

3.11 Highest common factor (booklet p. 23)

Examples (model these live)

E1 Find the HCF of 18 and 24.

18: 1, 2, 3, 6, 9, 18 · 24: 1, 2, 3, 4, 6, 8, 12, 24 · HCF = 6

E2 Find the HCF of 36 and 60 using prime factors.

$36 = 2^2 \times 3^2, 60 = 2^2 \times 3 \times 5$ · HCF = $2^2 \times 3 = 12$

E3 48 apples and 60 pears go into bags. Every bag is the same, with nothing left over. What is the largest number of bags?

HCF of 48 and 60 · = 12 bags

Questions (37 on today's skill, 16 from earlier lessons)

- | | | | |
|--|--|--|--|
| 1 3 | 18 6 | 30 $68\,000$ ^{3.02} | 42 common factors of all three · HCF = 8 |
| 2 no | 19 20: 1, 2, 4, 5, 10, 20 · 16: 1, 2, 4, 8, 16 · HCF = 4 | 31 HCF of 48 and 36 · = 12 groups | 43 short division, left to right $\cdot = 199$ ^{3.03} |
| 3 1, 2 | 20 HCF = $12 \cdot = 4/9$ | 32 HCF = $9 \cdot = 2/7$ | 44 common factors of all three · HCF = 6 |
| 4 3 | 21 $\approx 8\,000 \div 80 \cdot \approx 100$ ^{3.01} | 33 2, 3, 5 ^{3.10} | 45 HCF of 30 and 20 · = 10 cm squares |
| 5 no | 22 $63 = 3^2 \times 7, 28 = 2^2 \times 7 \cdot$ HCF = 7 | 34 $72 = 2^3 \times 3^2, 40 = 2^3 \times 5 \cdot$ HCF = 8 | 46 $180^\circ - 56^\circ = 124^\circ$ ^{2.09} |
| 6 1, 2 | 23 60: 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60 · 72: 1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72 · HCF = 12 | 35 e.g. $34 = 3 + 31$ ^{3.09} | 47 HCF of 30 and 40 · = 10 groups |
| 7 2 | 24 $= 7 + 9 = 16$ ^{3.08} | 36 HCF of 55 and 33 · = 11 groups | 48 22; no ^{3.04} |
| 8 no | 25 HCF = $36 \cdot = 2/3$ | 37 common factors of all three · HCF = 7 | 49 $33 = 3 \times 11, 55 = 5 \times 11 \cdot$ HCF = 11 |
| 9 $2^2 \times 3$ ^{3.10} | 26 HCF = $6 \cdot = 2/5$ | 38 $17 - (-15) = 17 + 15 = 32^\circ\text{C}$ ^{1.10} | 50 common factors of all three · HCF = 2 |
| 10 no | 27 $35 = 5 \times 7, 40 = 2^3 \times 5 \cdot$ HCF = 5 | 39 HCF of 30 and 50 · = 10 cm squares | 51 prime ^{3.09} |
| 11 1, 2, 3, 6 | 28 $\angle CYE$ or $\angle EYC$ ^{2.01} | 40 4 ^{3.08} | 52 HCF of 22 and 33 · = 11 cm squares |
| 12 8 | 29 78: 1, 2, 3, 6, 13, 26, 39, 78 · 65: 1, 5, 13, 65 · HCF = 13 | 41 HCF of 21 and 56 · = 7 groups | 53 $7 \times 7 \times 7 = 343 = 7$ ^{3.08} |
| 13 1 ^{1.03} | | | |
| 14 24: 1, 2, 3, 4, 6, 8, 12, 24 · 28: 1, 2, 4, 7, 14, 28 · HCF = 4 | | | |
| 15 no | | | |
| 16 1, 3 | | | |
| 17 $1 \times 27, 3 \times 9 \cdot 1, 3, 9, 27$ ^{3.09} | | | |

3.12 Lowest common multiple (booklet p. 25)

Examples (model these live)

E1 Find the LCM of 4 and 6.

4: 4, 8, 12 · 6: 6, 12 · LCM = 12

E2 Find the LCM of 8 and 12.

8: 8, 16, 24 · 12: 12, 24 · LCM = 24

E3 Buses leave every 12 min and every 18 min. Both leave at 8:00. When do they next leave together?

LCM of 12 and 18 = 36 · 8:36

Questions (38 on today's skill, 15 from earlier lessons)

- | | | | |
|---------------------|---------------------|-----------------------|---|
| 1 6 | 5 9, 18, 27, 36, 45 | 9 yes ^{3.11} | 13 6 ^{1.05} |
| 2 7, 14, 21, 28, 35 | 6 no | 10 6, 12, 18, 24, 30 | 14 10: 10, 20, 30 · 15: 15, 30 · LCM = 30 |
| 3 no | 7 24 | 11 yes | 15 3, 6, 9, 12, 15 |
| 4 15 | 8 8, 16, 24, 32, 40 | 12 12 | |

- 16 no
- 17 2×3^2 3.10
- 18 45
- 19 10: 10 · 5: 5, 10 · LCM = 10
- 20 list multiples of the largest; test the others · LCM = 12
- 21 $\approx 9\,000 \div 70 \cdot \approx 129$ 3.01
- 22 HCF = 1 · LCM = 182
- 23 9: 9, 18, 27, 36, 45 · 15: 15, 30, 45 · LCM = 45
- 24 71, 73, 79, 83, 89 3.09
- 25 list multiples of the largest; test the others · LCM = 12
- 26 HCF = 1 · LCM = 198
- 27 list multiples of the largest; test the others · LCM = 30
- 28 HCF = 1 · LCM = 247
- 29 $180^\circ - 62^\circ \cdot = 118^\circ$ 2.09
- 30 12: 12, 24, 36, 48, 60 · 15: 15, 30, 45, 60 · LCM = 60
- 31 37 000 3.02
- 32 LCM of 4 and 12 · = 12 seconds
- 33 list multiples of the largest; test the others · LCM = 30
- 34 1, 2, 4 3.11
- 35 HCF = 1 · LCM = 195
- 36 $= 2 \times 2 \times 2 \times 2 \times 3 \cdot = 48$ 3.10
- 37 LCM of 6 and 12 · = 12 seconds
- 38 LCM of 14 and 6 = 42 · 9:42
- 39 135° (check with a protractor) 2.02
- 40 LCM of 7 and 9 = 63 · 9 packs of hot dogs, 7 packs of buns
- 41 5 3.09
- 42 HCF = 1, LCM = 143 · 1 × 143 = 143 = 11 × 13
- 43 LCM of 5 and 9 · = 45 seconds
- 44 short division, left to right · = 179 3.03
- 45 LCM of 12 and 8 = 24 · 9:24
- 46 LCM of 5 and 12 = 60 · 12 packs of hot dogs, 5 packs of buns
- 47 -13 1.03
- 48 HCF = 1, LCM = 117 · 1 × 117 = 117 = 9 × 13
- 49 14; no 3.04
- 50 LCM of 6 and 7 · = 42 seconds
- 51 HCF = 3 · LCM = 36
- 52 $= 3 \times 7 \cdot = 21$ 3.10
- 53 LCM of 6 and 14 = 42 · 9:42